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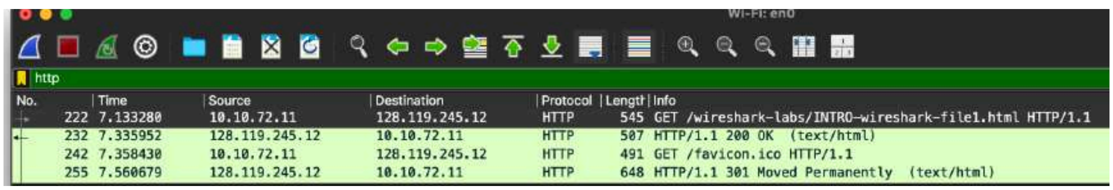
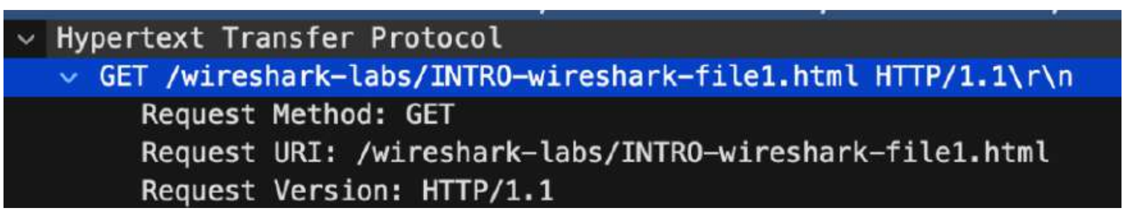
AIM: To use Wireshark to investigate the operational mechanics of the HTTP protocol, including message formats, conditional requests, and authentication. You will explore how browsers interact with servers to retrieve large files and embedded objects while managing connection persistence.

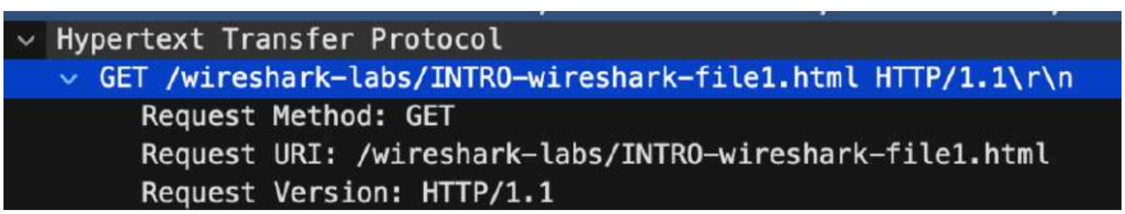
TASKS A, B, C, D, E

Q1] Is your browser running HTTP version 1.0 or 1.1? What version of HTTP is the server running?

PROOF:







ANS:

The browser is running HTTP/1.1, as shown in the request line:
GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1.
The server is also running HTTP/1.1, as shown in the response status line:
HTTP/1.1 200 OK.

Q2] What languages (if any) does your browser indicate that it can accept to the server? In the captured session, what other information (if any) does the browser provide the server with regarding the user/browser?

PROOF:

```
Host: gaia.cs.umass.edu\r\n
Connection: keep-alive\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Ch
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=
Accept-Encoding: gzip, deflate\r\n
Accept-Language: en-US,en;q=0.9\r\n
\r\n
```

ANS:

Accept-Language: en-US,en;q=0.9
Primary preference → US English (en-US)
Secondary preference → English (en) with slightly lower priority (q=0.9)

Q3] What languages (if any) does your browser indicate that it can accept to the server? In the captured session, what other information (if any) does the browser provide the server with regarding the user/browser?

PROOF:

```
Header Checksum: 0xc325 [validation disabled]
[Header checksum status: Unverified]
Source Address: 128.119.245.12
Destination Address: 10.10.72.11
[Stream index: 16]
```

ANS:

The IP address of my computer is 10.10.72.11 and the IP address of gaia.cs.umass.edu server is 128.119.245.12.

Q4]What is the status code returned from the server to your browser?

ANS:

STATUS CODE: 200
PHRASE: OK

Q5]When was the HTML file that you are retrieving last modified at the server?

PROOF:

```
> HTTP/1.1 200 OK\r\n
  Date: Tue, 17 Feb 2026 06:33:17 GMT\r\n
  Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.1 mod_fcgid/2.3.9 mod_perl/2.0.12 Pe
  Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n
  ETag: "51-64231b6715777"\r\n
```

ANS:

The HTML file was last modified on Tue, 28 Oct 2025 05:59:01 GMT.

Q6] How many bytes of content are being returned to your browser?

PROOF:

```
⌄ Content-Length: 81\r\n
    [Content length: 81]
```

ANS:

81 bytes of content are being returned to the browser.81 bytes of content are being returned to the browser.

Q7] By inspecting the raw data in the "packet bytes" pane, do you see any http headers within the data that are not displayed in the "packet details" pane? If so, name one.

PROOF:

```
0000  92 a4 01 a5 59 52 00 18  0a 4f 00 01 08 00 45 00   ····YR·· ·O····E·
0010  01 ed c9 4c 40 00 25 06  c3 25 80 77 f5 0c 0a 0a   ···L@·%· ·%·w····
0020  48 0b 00 50 e8 3a 1a 43  e0 c8 57 70 86 0a 80 18   H··P·:·C ··Wp····
0030  01 fa 8f 7c 00 00 01 01  08 0a 60 54 cb be b1 d0   ···|···· ··`T····
0040  9d 06 48 54 54 50 2f 31  2e 31 20 32 30 30 20 4f   ··HTTP/1 .1 200 O
0050  4b 0d 0a 44 61 74 65 3a  20 54 75 65 2c 20 31 37   K··Date:  Tue, 17
0060  20 46 65 62 20 32 30 32  36 20 30 36 3a 33 33 3a    Feb 202 6 06:33:
0070  31 37 20 47 4d 54 0d 0a  53 65 72 76 65 72 3a 20   17 GMT·· Server: 
0080  41 70 61 63 68 65 2f 32  2e 34 2e 36 32 20 28 41   Apache/2 .4.62 (A
0090  6c 6d 61 4c 69 6e 75 78  29 20 4f 70 65 6e 53 53   lmaLinux ) OpenSS
00a0  4c 2f 33 2e 35 2e 31 20  6d 6f 64 5f 66 63 67 69   L/3.5.1  mod_fcgi
00b0  64 2f 32 2e 33 2e 39 20  6d 6f 64 5f 70 65 72 6c   d/2.3.9  mod_perl
00c0  2f 32 2e 30 2e 31 32 20  50 65 72 6c 2f 76 35 2e   /2.0.12  Perl/v5.
00d0  33 32 2e 31 0d 0a 4c 61  73 74 2d 4d 6f 64 69 66   32.1··La st-Modif
00e0  69 65 64 3a 20 54 75 65  2c 20 32 38 20 4f 63 74   ied: Tue , 28 Oct
00f0  20 32 30 32 35 20 30 35  3a 35 39 3a 30 31 20 47    2025 05 :59:01 G
0100  4d 54 0d 0a 45 54 61 67  3a 20 22 35 31 2d 36 34   MT··ETag : "51-64
0110  32 33 31 62 36 37 31 35  37 37 37 22 0d 0a 41 63   231b6715 777"··Ac
0120  63 65 70 74 2d 52 61 6e  67 65 73 3a 20 62 79 74   cept-Ran ges: byt
0130  65 73 0d 0a 43 6f 6e 74  65 6e 74 2d 4c 65 6e 67   es··Cont ent-Leng
0140  74 68 3a 20 38 31 0d 0a  4b 65 65 70 2d 41 6c 69   th: 81·· Keep-Ali
0150  76 65 3a 20 74 69 6d 65  6f 75 74 3d 35 2c 20 6d   ve: time out=5, m
0160  61 78 3d 31 30 30 0d 0a  43 6f 6e 6e 65 63 74 69   ax=100·· Connecti
0170  6f 6e 3a 20 4b 65 65 70  2d 41 6c 69 76 65 0d 0a   on: Keep -Alive··
0180  43 6f 6e 74 65 6e 74 2d  54 79 70 65 3a 20 74 65   Content- Type: te
0190  78 74 2f 68 74 6d 6c 3b  20 63 68 61 72 73 65 74   xt/html;  charset
01a0  3d 55 54 46 2d 38 0d 0a  0d 0a 3c 68 74 6d 6c 3e   =UTF-8·· ··<html>
01b0  0a 43 6f 6e 67 72 61 74  75 6c 61 74 69 6f 6e 73   ·Congrat ulations
01c0  21 20 20 59 6f 75 27 76  65 20 64 6f 77 6e 6c 6f   !  You'v e downlo
01d0  61 64 65 64 20 74 68 65  20 66 69 72 73 74 20 57   aded the  first W
01e0  69 72 65 73 68 61 72 6b  20 6c 61 62 20 66 69 6c   ireshark  lab fil
01f0  65 21 0a 3c 2f 68 74 6d  6c 3e 0a                  e!·</htm l>·
```

ANS:

Yes, one HTTP header visible in the raw packet bytes is "Accept-Ranges: bytes".

Q8] Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?

PROOF:

```
Hypertext Transfer Protocol
  GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1\r\n
    Request Method: GET
    Request URI: /wireshark-labs/INTRO-wireshark-file1.html
    Request Version: HTTP/1.1
  Host: gaia.cs.umass.edu\r\n
  Connection: keep-alive\r\n
  Upgrade-Insecure-Requests: 1\r\n
  User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chr
  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0
  Accept-Encoding: gzip, deflate\r\n
  Accept-Language: en-US,en;q=0.9\r\n
  \r\n
  [Response in frame: 232]
  [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html]
Text item (text), 57 bytes                                                              Packets:
```

ANS:

No, the first HTTP GET request does not contain an If-Modified-Since header. It is not visible because this is the initial request, and conditional headers are only included in subsequent requests after the object has been cached.

Q9]Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?

PROOF:

```
Line-based text data: text/html (3 lines)
  <html>\n
  Congratulations!  You've downloaded the first Wireshark lab file!\n
  </html>\n
```

ANS: Yes, it did.

Q10]Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE:" line in the HTTP GET? If so, what information follows the "IF-MODIFIED-SINCE:" header?

PROOF:

```
⌄ Hypertext Transfer Protocol
  › GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
    Cache-Control: max-age=0\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
    Accept: text/html,application/xhtml+xml,application/xml;q=0
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    If-None-Match: "51-64231b6715777"\r\n
    If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT\r\n
```

ANS:

Yes, the second HTTP GET request contains an If-Modified-Since header.

The value following it is: Tue, 28 Oct 2025 05:59:01 GMT.

Q11]What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

PROOF:

```
⌄ Hypertext Transfer Protocol
  › HTTP/1.1 304 Not Modified\r\n
    Date: Tue, 17 Feb 2026 07:17:24 GMT\r\n
    Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.1 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1\r\n
    Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n
    ETag: "51-64231b6715777"\r\n
    Accept-Ranges: bytes\r\n
    Keep-Alive: timeout=5, max=100\r\n
    Connection: Keep-Alive\r\n
    \r\n
    [Request in frame: 565]
    [Time since request: 216.936000 milliseconds]
    [Request URI: /wireshark-labs/INTRO-wireshark-file1.html]
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html]
```

ANS:

The HTTP status code and phrase returned is 304 Not Modified.

No, the server did not explicitly return the contents of the file. This is because a 304 Not Modified response means the cached version in the browser is still valid, so the server does not send the file body again.

Q12]How many HTTP GET request messages were sent by your browser?

PROOF:

No.	Time	Source	Destination	Protocol	Length	Info
284	6.447956	10.10.72.11	128.119.245.12	HTTP	544	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
307	6.662427	128.119.245.12	10.10.72.11	HTTP	586	HTTP/1.1 200 OK (text/html)
311	6.695394	10.10.72.11	128.119.245.12	HTTP	490	GET /favicon.ico HTTP/1.1
332	6.907506	128.119.245.12	10.10.72.11	HTTP	648	HTTP/1.1 301 Moved Permanently (text/html)

ANS:

Based on the Wireshark packet capture, there were 2 HTTP GET request messages sent by the browser. The first request is found in packet number 284, where the browser requests the file /wireshark-labs/HTTP-wireshark-file3.html. The second request is found in packet number 311, where the browser automatically attempts to retrieve the website's icon file, /favicon.ico

Q13]How many data-containing TCP segments were needed to carry the single HTTP response?

PROOF:

```
> Transmission Control Protocol, Src Port: 80, Dst Port: 59648, Seq: 4345, Ack: 479, Len: 520
> [4 Reassembled TCP Segments (4864 bytes): #304(1448), #305(1448), #306(1448), #307(520)]
```

ANS:

There were 4 data-containing TCP segments needed to carry the single HTTP response for the HTML file. Wireshark indicates this in the packet details under the field [4 Reassembled TCP Segments], showing that the data was split across frames 304, 305, 306, and 307 to accommodate the total message size of 4864 bytes.

Q14]What is the status code and phrase associated with the response to the HTTP GET request?

ANS:

STATUS CODE: 200

PHRASE: OK

Q15] Is there any HTTP header information in the transmitted data associated with TCP segmentation?

ANS:
There is no HTTP header information associated with TCP segmentation. Segmentation is a Transport Layer (TCP) process designed to break data into segments that fit within the network's Maximum Segment Size (MSS). Since HTTP is an Application Layer protocol, it is unaware of this process. The information you see in Wireshark, such as "[4 Reassembled TCP Segments]," is an analysis metadata field added by Wireshark to track frames 304, 305, 306, and 307, and is not an actual header transmitted in the packet data.

Q16]How many HTTP GET request messages were sent by your browser? To which Internet addresses were these GET requests sent?

PROOF:

No.	Time	Source	Destination	Protoco	Lengt	Info
125	4.180763	10.10.72.11	128.119.245.12	HTTP	544	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
142	4.392129	128.119.245.12	10.10.72.11	HTTP	13…	HTTP/1.1 200 OK (text/html)
144	4.411761	10.10.72.11	128.119.245.12	HTTP	490	GET /pearson.png HTTP/1.1
191	4.622356	128.119.245.12	10.10.72.11	HTTP	648	HTTP/1.1 301 Moved Permanently (text/html)
206	4.640054	10.10.72.11	2.56.99.24	HTTP	457	GET /8E_cover_small.jpg HTTP/1.1
632	5.477110	2.56.99.24	10.10.72.11	HTTP	12…	HTTP/1.1 200 OK (JPEG JFIF image)
634	5.481283	10.10.72.11	128.119.245.12	HTTP	490	GET /favicon.ico HTTP/1.1
638	5.691989	128.119.245.12	10.10.72.11	HTTP	648	HTTP/1.1 301 Moved Permanently (text/html)

ANS:

Based on the packet captures, there were 4 unique GET requests sent by the browser. These include requests for file4.html (Frame 125), pearson.png (Frame 144), 8E_cover_small.jpg (Frame 206), and favicon.ico (Frame 634).

To which Internet addresses were these GET requests sent? The GET requests were sent to two different IP addresses: 128.119.245.12 and 2.56.99.24. Specifically, the HTML and icon files were requested from 128.119.245.12, while the image file 8E_cover_small.jpg was requested from 2.56.99.24.

Q17]Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain.

ANS:

The browser downloaded the two images in parallel. As seen in the packet capture, the GET request for pearson.png (Frame 144) was sent at 4.411761 seconds to 128.119.245.12, and the GET request for 8E_cover_small.jpg (Frame 206) was sent at 4.640054 seconds to

2.56.99.24. The browser did not wait for the response to the first image request before initiating the second request to a different server, which is a standard behavior in modern browsers to optimize page loading times by utilizing multiple simultaneous TCP connections.

Q18]What is the server's response (status code and phrase) in response to the initial HTTP GET message from your browser?

PROOF:

No.	Time	Source	Destination	Protoco	Lengt	Info
275	11.943126	10.10.72.11	128.119.245.12	HTTP	560	GET /wireshark-labs/protected_pages/HTTP-
287	12.145761	128.119.245.12	10.10.72.11	HTTP	786	HTTP/1.1 401 Unauthorized (text/html)
2963	135.312421	10.10.72.11	128.119.245.12	HTTP	645	GET /wireshark-labs/protected_pages/HTTP-
2968	135.573702	128.119.245.12	10.10.72.11	HTTP	559	HTTP/1.1 200 OK (text/html)
2972	135.666612	10.10.72.11	128.119.245.12	HTTP	506	GET /favicon.ico HTTP/1.1
2974	135.982244	128.119.245.12	10.10.72.11	HTTP	648	HTTP/1.1 301 Moved Permanently (text/ht

ANS:

STATUS CODE: 401

PHRASE: UNAUTHORIZED

Q19]When your browser sends the HTTP GET message for the second time, what new field is included in the HTTP GET message?

PROOF:

No.	Time	Source	Destination	Protoco	Lengt	Info
275	11.943126	10.10.72.11	128.119.245.12	HTTP	560	GET /wireshark-labs/protected_pages/HTTP-
287	12.145761	128.119.245.12	10.10.72.11	HTTP	786	HTTP/1.1 401 Unauthorized (text/html)
2963	135.312421	10.10.72.11	128.119.245.12	HTTP	645	GET /wireshark-labs/protected_pages/HTTP-
2968	135.573702	128.119.245.12	10.10.72.11	HTTP	559	HTTP/1.1 200 OK (text/html)
2972	135.666612	10.10.72.11	128.119.245.12	HTTP	506	GET /favicon.ico HTTP/1.1
2974	135.982244	128.119.245.12	10.10.72.11	HTTP	648	HTTP/1.1 301 Moved Permanently (text/ht

ANS:
When the browser sends the HTTP GET message for the second time (Frame 2963), the new field included is Authorization. This field contains the credentials (username and password) required to access the protected page after the server initially rejected the first request with a "401 Unauthorized" status.

Q20]What does the "Connection: close" and "Connection: Keep-alive" header field imply in HTTP protocol? When should one be used over the other?
ANS:

Connection: Keep-alive This implies a persistent connection. The TCP connection stays open after the response is sent, allowing the browser to send multiple HTTP requests (like fetching images, CSS, and JS files) over the same single connection.

Connection: close This implies a non-persistent connection. The server or client informs the other party that the TCP connection will be terminated immediately after the current request/response cycle is finished.

When to use each:

- Keep-alive: Use this for modern web pages that contain multiple objects (images, scripts). It reduces latency and CPU usage because it avoids the overhead of the "three-way handshake" required to open a new TCP connection for every single item.
- close: Use this for one-off requests where no further data is expected, or on high-traffic servers to free up memory and connection slots by preventing idle connections from staying open.

Conclusion:
This experiment demonstrated the mechanics of the HTTP protocol by analyzing packet captures of various browser-server interactions. Through these observations, the fundamental request-response cycle, the use of conditional GETs for caching efficiency, and the handling of multi-segment TCP transmissions for large files were verified. Furthermore, the analysis of embedded objects and authentication revealed how browsers optimize performance through parallel downloads and handle security via header-based credential encoding.